

Detailed Balance Normalizing Flow for Generative Sampling

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1 Introduction

Normalizing flows based on the KL divergence (Rezende and Mohamed, 2015) have been widely applied in generative sampling. In energy-based sampling from physical potentials, the Boltzmann generator (Noé et al., 2019; Wu et al., 2020) has been the industry standard for fast sampling from target distributions. For data-driven sampling problems, although normalizing flows are less popular than score-based diffusion (Ho et al., 2020; Song et al., 2021) and flow matching (Lipman et al., 2023), they remain a versatile tool across a range of applications, including density estimation (Papamakarios et al., 2017; Kingma and Dhariwal, 2018), simulation-based inference (Papamakarios et al., 2019), and anomaly detection (Kirichenko et al., 2020; Rudolph et al., 2021).

Despite their popularity, KL normalizing flows suffer from several intrinsic drawbacks. First, the flow map is a diffeomorphism on the whole space, so for multi-modal distributions the resulting NF often introduces artificial bridges between modes. Second, the KL loss is highly sensitive to mode collapse, and the flow easily falls into a few dominant modes while neglecting the rest.

In this paper, we propose a normalizing flow based on the detailed balance (DB) divergence, which serves as an alternative for multi-modal generative sampling tasks. Its key advantage is that the DB divergence matches the gradient rather than the function value and keeps the loss function nonnegative. This incurs only a constant-factor overhead in training while offering substantially more flexibility in use.

2 Detailed balance divergence

2.1 Basic definition

We introduce the detailed balance (DB) divergence to characterize the discrepancy between two distributions. Let μ and ν be probability distributions on $\mathbb{R}^d$. The DB divergence is defined as

$$\text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)| dx = \int_{\mathbb{R}^d} \mu(x) \left| \nabla \left(\frac{\nu(x)}{\mu(x)} \right) \right| dx, \quad (1)$$

which is reminiscent of both the KL divergence

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx,$$

and the Fisher information

$$\text{Fisher}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)|^2 dx.$$

By definition, $\text{DB}(\nu\|\mu)$ is nonsymmetric and nonnegative, and it vanishes precisely when the ratio $\nu(x)/\mu(x)$ is constant, that is, when $\mu = \nu$.

If μ takes the unnormalized form $\mu(x) \propto e^{-U(x)}$ with potential $U(x)$, then

$$\text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} |\nabla \nu(x) + \nu(x) \nabla U(x)| dx,$$

so minimizing the DB divergence amounts to solving the PDE

$$\nabla \nu(x) + \nu(x) \nabla U(x) = 0, \quad (2)$$

which is exactly the detailed balance condition for sampling $e^{-U(x)}$ via Langevin dynamics. The DB divergence therefore carries a natural physical meaning.

2.2 Relations to other distribution metrics

The DB divergence is closely related to two classical distribution metrics, the Fisher information and the total variation distance, and sits naturally between the two in strength.

Fisher information. The Fisher information of ν relative to μ is

$$\text{Fisher}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)|^2 dx,$$

so by (1) the DB divergence is precisely its L^1 counterpart, with the squared L^2 norm of the score difference replaced by an L^1 norm. The Cauchy–Schwarz inequality together with $\int \nu(x) dx = 1$ then yields

$$\text{DB}(\nu\|\mu)^2 \leq \text{Fisher}(\nu\|\mu). \quad (3)$$

Total variation distance. The total variation distance between μ and ν is

$$\text{TV}(\mu, \nu) = \frac{1}{2} \int_{\mathbb{R}^d} |\mu(x) - \nu(x)| dx.$$

Setting $f(x) = \nu(x)/\mu(x) - 1$, so that $\int \mu(x) f(x) dx = 0$ since μ and ν are normalized, we obtain

$$\text{TV}(\mu, \nu) = \frac{1}{2} \int_{\mathbb{R}^d} \mu(x) |f(x)| dx, \quad \text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} \mu(x) |\nabla f(x)| dx.$$

If μ satisfies a Cheeger inequality (an L^1 Poincaré inequality) with isoperimetric constant $c_P > 0$, applying it to the mean-zero function f yields

$$\text{TV}(\mu, \nu) \leq \frac{1}{2c_P} \text{DB}(\nu\|\mu). \quad (4)$$

Such a Cheeger inequality holds for log-concave distributions and for Gaussian mixtures with sufficient overlap, so the DB divergence controls the total variation distance for these classes.

Combining (3) and (4) yields the hierarchy

$$\text{TV}(\mu, \nu) \leq \frac{1}{2c_P} \text{DB}(\nu\|\mu) \leq \frac{1}{2c_P} \sqrt{\text{Fisher}(\nu\|\mu)}, \quad (5)$$

placing the DB divergence between the total variation distance and the Fisher information. It is weaker than the Fisher information, yet still strong enough to control total variation under regularity assumptions on μ .

To illustrate the hierarchy (5), consider the one-dimensional family

$$\mu_\sigma \propto e^{-U_\sigma(x)}, \quad U_\sigma(x) = \frac{1}{2}x^2 + \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2\sigma^2}}, \quad \nu \propto e^{-V(x)}, \quad V(x) = \frac{1}{2}x^2,$$

indexed by $\sigma \in \{2^0, 2^{-1}, \dots, 2^{-7}\}$. The potentials U_σ and V differ by a Gaussian-shaped bump whose total mass equals σ , so as $\sigma \rightarrow 0$ the perturbation becomes thinner and taller but carries less and less probability mass. Figure 1 shows that the three metrics react to this in very different ways: KL decays at the rate of the bump’s mass and effectively ignores the perturbation; DB treats it as a fixed-size disturbance and stays roughly constant; Fisher amplifies the steep gradient and diverges.

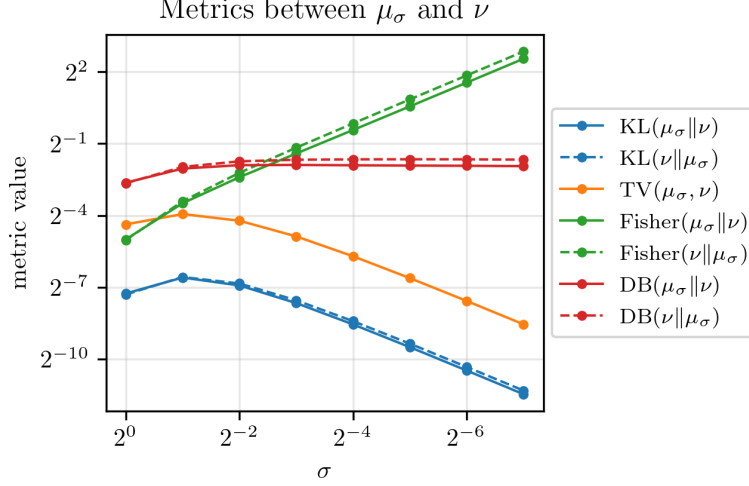


Figure 1: The four metrics between μ_σ and ν for $\sigma = 2^0, 2^{-1}, \dots, 2^{-7}$. KL and TV decay with σ ; DB stays roughly constant; Fisher diverges.

3 Energy-based DB normalizing flow

Let $U_0(x)$ and $U_1(x)$ be source and target potential functions on $\mathbb{R}^d$, and let $\mu_0(x) \propto \exp(-U_0(x))$ and $\mu_1(x) \propto \exp(-U_1(x))$ be the corresponding Boltzmann distributions. Given explicit samples from μ_0 , we aim to train a flow map F such that $F_{\#}\mu_0 = \mu_1$, with pushforward density

$$(F_{\#}\mu_0)(y) = \frac{\mu_0(x)}{|\det J_F(x)|}, \quad y = F(x), \quad (6)$$

where $J_F(x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of F . We construct the loss from the DB divergence between $F_{\#}\mu_0$ and μ_1 in the reverse direction, $DB(F_{\#}\mu_0 || \mu_1)$, which after a change of variable becomes an integral against the prior and can be estimated without any samples from the target.

3.1 Construction of reverse DB loss

First, we define the log importance weight

$$W_F(x) = \log \frac{(F_{\#}\mu_0)(F(x))}{\mu_1(F(x))} + \text{const} = U_1(F(x)) - U_0(x) - \log |\det J_F(x)|. \quad (7)$$

Viewing $W_F(x)$ as a function of $y = F(x)$, we have

$$\nabla_y W_F(x) = \nabla \log(F_{\#}\mu_0)(y) - \nabla \log \mu_1(y).$$

Applying the DB divergence in (1) and the change of variable $y = F(x)$,

$$DB(F_{\#}\mu_0 || \mu_1) = \int_{\mathbb{R}^d} (F_{\#}\mu_0)(y) |\nabla_y W_F(x)| dy = \int_{\mathbb{R}^d} \mu_0(x) |\nabla_y W_F(x)| dx. \quad (8)$$

The gradient $\nabla_y W_F$ admits an explicit chain-rule expansion. Differentiating (7) with respect to $y = F(x)$,

$$\begin{aligned} \nabla_y W_F(x) &= \nabla U_1(F(x)) - \nabla_y (U_0(x) + \log |\det J_F(x)|) \\ &= \nabla U_1(F(x)) - J_F(x)^{-\top} (\nabla U_0(x) + \nabla \log |\det J_F(x)|). \end{aligned} \quad (9)$$

Substituting (9) into (8), the loss in terms of the flow map F takes the form

$$DB(F_{\#}\mu_0 || \mu_1) = \int_{\mathbb{R}^d} \mu_0(x) \left| \nabla U_1(F(x)) - J_F(x)^{-\top} (\nabla U_0(x) + \nabla \log |\det J_F(x)|) \right| dx. \quad (10)$$

The loss (10) is highly nonlinear in F , primarily because of the explicit appearance of the Jacobian matrix $J_F(x)$ rather than only its determinant. Standard normalizing flow architectures such as RealNVP and NSF support efficient evaluation of the determinant, but assembling the full Jacobian costs at least $O(d^2)$, which becomes prohibitive in high dimensions even with automatic differentiation.

To circumvent this, we bypass the chain-rule expansion (9) altogether and directly approximate the directional derivative $v^\top \nabla_y W_F(x)$ along a random direction $v \in \mathbb{S}^{d-1}$ by central differences in $y = F(x)$. Fix a small step size $\varepsilon > 0$ and define

$$x_{v\pm} = F^{-1}(F(x) \pm \varepsilon v), \quad (11)$$

so that $F(x_{v\pm}) = F(x) \pm \varepsilon v$. Evaluating W_F from (7) at $x_{v\pm}$ and central-differencing in y gives

$$v^\top \nabla_y W_F(x) \approx \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon}, \quad (12)$$

which by (7) expands to

$$\begin{aligned} \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} &= \frac{U_1(F(x) + \varepsilon v) - U_1(F(x) - \varepsilon v)}{2\varepsilon} \\ &\quad - \frac{(U_0(x_{v+}) + \log |\det J_F(x_{v+})|) - (U_0(x_{v-}) + \log |\det J_F(x_{v-})|)}{2\varepsilon}. \end{aligned} \quad (13)$$

The right-hand side requires only forward evaluations of U_1 , F^{-1} , U_0 , and $\log |\det J_F|$ — all available cheaply in NSF-style architectures — and never assembles $J_F(x)$ itself. The reverse DB loss (8) is finally estimated by averaging $|v^\top \nabla_y W_F(x)|$ over $x \sim \mu_0$ and $v \sim \text{Unif}(\mathbb{S}^{d-1})$:

$$\mathcal{L}_{\text{reverse}}[F] = \mathbb{E}_{\substack{x \sim \mu_0 \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|. \quad (14)$$

Remark *The directional-derivative trick is well-matched to the reverse DB loss but not to a Fisher-based variant. For $v \sim \text{Unif}(\mathbb{S}^{d-1})$ and any $g \in \mathbb{R}^d$,*

$$\mathbb{E}_v |v^\top g| = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi} \Gamma(\frac{d+1}{2})} |g| \sim \sqrt{\frac{2}{\pi d}} |g|, \quad \mathbb{E}_v (v^\top g)^2 = \frac{1}{d} |g|^2,$$

where the asymptotic on the left follows from Stirling’s formula. The DB integrand is $|g| = |\nabla_y W_F|$, so $O(\sqrt{d})$ random directions recover the precision of one full-gradient evaluation, costing roughly $\sqrt{d}$ times a single backward pass. The Fisher information involves $|g|^2$, for which $O(d)$ random directions are needed to match a full gradient — comparable to assembling $J_F(x)$ explicitly. The directional-derivative trick therefore makes sense for the DB normalizing flow but offers no asymptotic saving for its Fisher analogue.

3.2 Mode guidance with forward DB loss

Similar to the reverse DB loss, the forward DB loss $\text{DB}(\mu_1 \| F_\# \mu_0)$ can be written as

$$\text{DB}(\mu_1 \| F_\# \mu_0) = \int_{\mathbb{R}^d} \mu_1(y) |\nabla_y W_F(x)| dy, \quad (15)$$

the only difference from (8) being that the integration measure is the target μ_1 rather than the prior μ_0 . The same central-difference scheme as in (12) applies: for a fixed $y \in \mathbb{R}^d$, set

$$x_{v\pm} = F^{-1}(y \pm \varepsilon v), \quad (16)$$

so that $F(x_{v\pm}) = y \pm \varepsilon v$, and approximate $v^\top \nabla_y W_F(x)$ at $x = F^{-1}(y)$ by

$$\begin{aligned} \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} &= \frac{U_1(y + \varepsilon v) - U_1(y - \varepsilon v)}{2\varepsilon} \\ &\quad - \frac{(U_0(x_{v+}) + \log |\det J_F(x_{v+})|) - (U_0(x_{v-}) + \log |\det J_F(x_{v-})|)}{2\varepsilon}, \end{aligned} \quad (17)$$

yielding the forward loss

$$\mathcal{L}_{\text{forward}}[F] = \mathbb{E}_{\substack{y \sim \mu_1 \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|. \quad (18)$$

The only difference from (14) is that y is drawn from the target μ_1 , so the computation graph tracks y rather than $x = F^{-1}(y)$.

Remark *Unlike the KL divergence, which is invariant under pushforward by a diffeomorphism, the DB divergence is not; in general*

$$\text{DB}(\mu_1 \| F_{\#}\mu_0) \neq \text{DB}(F_{\#}^{-1}\mu_1 \| \mu_0),$$

because a change of variable transforms the score difference by the Jacobian factor $J_F^{\top}$, whose L^1 norm is not preserved unless F is an isometry. In principle, $\text{DB}(F_{\#}^{-1}\mu_1 \| \mu_0)$ could also serve as a forward DB loss, but we keep $\text{DB}(\mu_1 \| F_{\#}\mu_0)$ because its derivation is simpler.

In general the forward loss $\mathcal{L}_{\text{forward}}[F]$ in (18) cannot be evaluated directly, since it requires explicit samples from the target μ_1 . The crucial difference between the DB and KL losses is that the DB integrand $|\nabla_y W_F(x)|$ is pointwise nonnegative: replacing μ_1 in (18) by an easier-to-sample guidance distribution $\hat{\mu}_1$ yields a loss $\hat{\mathcal{L}}_{\text{forward}}[F]$ that still vanishes exactly when $F_{\#}\mu_0 \propto \mu_1$ on $\text{supp}(\hat{\mu}_1)$ — the global minimizer is preserved. KL has no such freedom: its integrand $\log(\mu_1/F_{\#}\mu_0)$ is signed, so any inaccuracy in μ_1 shifts the optimum to $F_{\#}\mu_0 = \hat{\mu}_1$, and KL training therefore demands exact samples from μ_1 . For DB it suffices to pick $\hat{\mu}_1$ to cover the modes the reverse loss tends to drop. We take $\hat{\mu}_1$ to be an unnormalized Gaussian mixture

$$\hat{\mu}_1(y) = \sum_{i=1}^n w_i \mathcal{N}(y; x_i, \Sigma_i), \quad w_i \geq 0, \quad (19)$$

where $\{x_i\}_{i=1}^n$ are the local minima of U_1 and $\{\Sigma_i\}_{i=1}^n$ are user-chosen covariances; the weights w_i tune the strength of mode guidance per mode and absorb the role formerly played by a global multiplier. Linearity then expands $\hat{\mathcal{L}}_{\text{forward}}[F]$ as a per-component sum,

$$\hat{\mathcal{L}}_{\text{forward}}[F] = \sum_{i=1}^n w_i \mathbb{E}_{\substack{y \sim \mathcal{N}(x_i, \Sigma_i) \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|, \quad (20)$$

with $x_{v\pm}$ as in (16). The total loss is then simply

$$\mathcal{L}_{\text{total}}[F] = \mathcal{L}_{\text{reverse}}[F] + \hat{\mathcal{L}}_{\text{forward}}[F]. \quad (21)$$

The guidance term supplies mode information from the target side and effectively suppresses the mode collapse that pure energy-based normalizing flows commonly suffer from.

3.3 Construction of Boltzmann generator

4 Detailed Balance Score Matching

Although DBNF cannot be directly used for generative sampling, it can be combined with standard approaches such as score matching. Suppose that samples from a multi-modal target distribution $\mu(x)$ are available, and we seek a fast sampling method. As before, we choose μ_0 as a tractable adaptive prior and train a flow map F to minimize the reverse DB loss $\text{DB}(F_{\#}\mu_0 \| \mu)$. Once μ_0 and F are trained, the distributions μ_0 and $F_{\#}\mu_0$ are close, and the remaining gap can be bridged by a standard generative layer such as score-based diffusion, which is able to overcome the geometric constraints of diffeomorphisms.

The central challenge remains the evaluation of the reverse DB loss (10). In this setting, samples from $\mu(x)$ are available, so importance sampling is not needed. The main difficulty is that the gradient $\nabla U(x)$ is not directly accessible from samples alone.

Nevertheless, we can employ implicit score matching to obtain an approximation to $\nabla U(x)$. Let $s(x) \in \mathbb{R}^d$ denote a parametric score model. The implicit score matching loss is

$$\int_{\mathbb{R}^d} \mu(x) |s(x) - \nabla U(x)|^2 dx = \int_{\mathbb{R}^d} \mu(x) (|s(x)|^2 + 2\nabla \cdot s(x)) dx, \quad (22)$$

where the right-hand side depends only on samples from $\mu(x)$ and does not require explicit knowledge of $U(x)$. Minimizing (22) yields $s(x) \approx \nabla U(x)$, which can then be substituted into the reverse DB loss (10). With this learned gradient, we train the flow map F and the adaptive prior $\mu_0(x)$ as in DBNF. Finally, a score-based diffusion layer is added between μ_0 and $F_{\#}\mu$ to close the remaining gap. Since μ_0 and $F_{\#}\mu$ are already close after DBNF training, the diffusion process only needs to run for a short time, which is expected to yield efficient and high-quality generative sampling.

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